



Submitted By:

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Assignment:

Lab No. 3

Submitted To:

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For the fulfillment of,

Object-Oriented Programming Lab

Department of Computer Engineering

University of Engineering and Technology, Lahore

TASK 1:

2-D to 3-D:

CODE:

```
import java.lang.Math;

class point {

    private int x;

    private int y;

    private int z;


    public int getX() {

        return x;

    }


    public void setX(int x) {

        this.x = x;

    }


    public int getY() {

        return y;

    }


    public void setY(int y) {

        this.y = y;

    }


    public int getZ() {

        return z;

    }


    public void setZ(int z) {

        this.z = z;

    }

}
```

```

    }

    public double distance (point a) {
        double dis = Math.sqrt((x - a.x) * (x - a.x) + (y - a.y) * (y - a.y) + (z - a.z) * (z - a.z));
        return dis;
    }
}

```

```

public class lab3_t1 {
    public static void main(String[] args) {
        point p1 = new point();
        point p2 = new point();

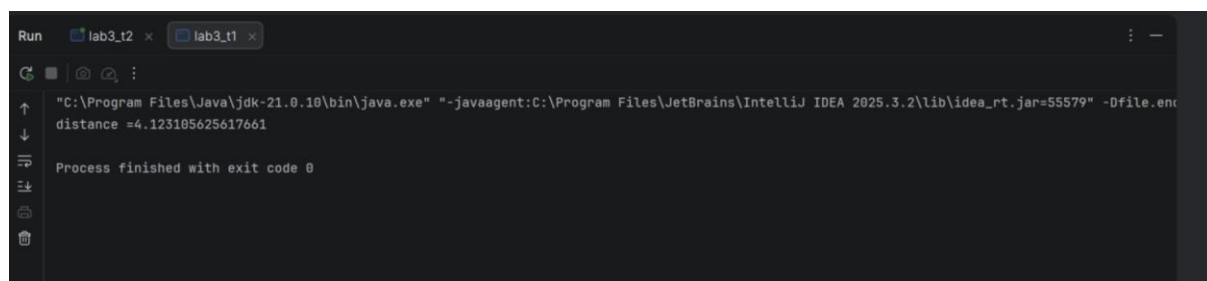
        p1.setX(4);
        p1.setY(2);
        p1.setZ(9);

        p2.setX(0);
        p2.setY(2);
        p2.setZ(8);

        double distance = p1.distance(p2);
        System.out.println("distance =" + distance);
    }
}

```

OUTPUT:



```

Run  lab3_t2 x  lab3_t1 x
C:\Program Files\Java\jdk-21.0.10\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=55579" -Dfile.encoding=UTF-8
distance =4.123185625617661
Process finished with exit code 0

```

TASK 2:

Time:

CODE:

```
class Time {  
  
    private int hours;  
    private int minutes;  
    private int seconds;  
  
    public int getSeconds() {  
        return seconds;  
    }  
  
    public void setSeconds(int seconds) {  
        this.seconds = seconds;  
    }  
  
    public int getMinutes() {  
        return minutes;  
    }  
  
    public void setMinutes(int minutes) {  
        this.minutes = minutes;  
    }  
  
    public int getHours() {  
        return hours;  
    }  
  
    public void setHours(int hours) {
```

```

        this.hours = hours;
    }

    public void whatTime(int no_of_seconds_till_noon) {

        int totalSeconds = 43200 - no_of_seconds_till_noon;

        hours = totalSeconds / 3600;
        minutes = (totalSeconds % 3600) / 60;
        seconds = totalSeconds % 60;
    }
}

public class lab3_t2 {

    public static void main(String[] args) {

        Time t = new Time();
        t.whatTime(3600);

        int h = t.getHours();
        int m = t.getMinutes();
        int s = t.getSeconds();

        System.out.println("hours= " + h + "\nminutes= " + m + "\nseconds= " + s);

    }
}

```

OUTPUT:



```

C:\Program Files\Java\jdk-21.0.10\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=55661" -Dfile.encoding=UTF-8
hours= 11
minutes= 0
seconds= 0
Process finished with exit code 0

```

TASK 3:

Down payment:

CODE:

```
import java.util.Scanner;

public class lab3_t2 {

    public static void main(String[] args) {

        Scanner sc= new Scanner(System.in);

        System.out.println("Enter the cost of your dream house :");

        long total_cost= sc.nextLong();

        System.out.print("Enter the decimal of salary to you want to save :");

        float portion_saved = sc.nextFloat();

        System.out.print("Enter your monthly salary :");

        long monthly_salary = sc.nextLong();

        double portion_down_payment = 0.25 * total_cost;

        double r = 0.04;

        double current_savings = 0;

        int months = 0;

        while (current_savings < portion_down_payment) {

            current_savings += current_savings * r / 12;

            current_savings += monthly_salary * portion_saved;

            months++;

        }

        System.out.println("Number of months =" + months);
```

```
}  
  
}
```

OUTPUT:

```
"C:\Program Files\Java\jdk-21.0.10\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=55549" -Dfile.encoding=UTF-8  
Enter the cost of your dream house :  
200000  
Enter the decimal of salary to you want to save :0.2  
Enter your monthly salary :40000  
Number of months =7  
  
Process finished with exit code 0  
|
```

```
"C:\Program Files\Java\jdk-21.0.10\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=55569" -Dfile.encoding=UTF-8  
Enter the cost of your dream house :  
1000000  
Enter the decimal of salary to you want to save :0.1  
Enter your monthly salary :1000  
Number of months =672  
  
Process finished with exit code 0  
|
```